

## Xinyun Liu, PhD

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### SUMMARY

Machine Learning Scientist with 4+ years of experience specializing in AI security. Hands-on experience implementing core ML components from scratch and deploying containerized services on AWS (EC2, Docker). Strong foundations in LLM systems, RAG architectures, and ML system design.

### SKILLS

Python, C/C++, Java, Matlab, LATEX, Scikit-Learn, Numpy, Scipy, Pandas, Matplotlib, Spark, Pytorch, Tensorflow, Keras, Flask, Git, Github, GCP, SQL, Docker, Linux/Unix, Flask, Streamlit, AWS (EC2), Kubernetes, NLP, LLM Post-training, RAG, SFT (LoRA), RLHF, LangChain, Transformer, Prompt Engineering, CNNs, GANs, Diffusion Models, ResNet, U-Net, GNN, ZKP, Blockchain, Linear Algebra.

### EXPERIENCE

#### Michigan Technological University

Houghton, MI

##### Graduate Research Assistant

01/2022 - Present

##### *End-to-end Watermarking System*

- Designed and implemented a dual-layer watermarking framework for federated learning model provenance and data integrity verification, achieving 99% detection accuracy against data manipulation and poisoning attacks, reducing attack success rate from 100% to 5% while preserving model utility within 96% of baseline accuracy.
- Built end-to-end PyTorch training and evaluation pipelines across 5 benchmark datasets, supporting reproducible experiments over 6 model architectures. Optimized inference and evaluation throughput by 32% via batched evaluation and GPU profiling, enabling large-scale ablation studies.

##### *Coronary Artery Segmentation*

- Developed a novel 3D deep learning network (AGFA-Net) for automatic segmentation of coronary arteries from 3D CCTA images.
- Integrated attention-guided feature refinement, scale-adaptive feature augmentation, and hierarchical feature integration to address low contrast and complex anatomical structures, achieving a Dice score of 86.74% and an HD95 of 0.82 mm on a large-scale dataset.

##### *Automatic Restoration of Poisoned Training Data*

- Developed PD-AutoR (PMLR 2025), a trigger-agnostic defense framework that automatically detects and restores poisoned training samples instead of discarding them, preserving scarce training data while mitigating data poisoning attacks.
- Built the end-to-end pipeline: a fusion-feature detector (post-training features + training-loss-bifurcation signals) with SVD-based outlier detection, and a transductive cycleGAN restorator using ViT-aided U-Net generators and a discriminator.
- Achieved up to 98.5% label restoration accuracy and restoration error (APD) as low as 0.007 across 4 benchmarks and 4 attack types; the multi-granularity design cut restoration error  $\sim 8\times$  (APD 0.192 $\rightarrow$ 0.023); downstream model accuracy consistently beat the discard baseline (up to +1.23pp) with fault tolerance to 8% detection error.

#### Shenzhen Jinghong Technology Limited

Shenzhen, Guangdong, China

##### Machine Learning Engineer (3D Vision)

08/2019 - 08/2021

##### *3D ToF Depth Sensing & Gesture Recognition*

- Core developer of a Time-of-Flight (ToF) 3D camera system for indoor scene reconstruction, achieving 1% point-cloud depth accuracy at 5m range; product commercialized and delivered to customers.
- Developed a depth-based gesture recognition module for automotive applications, enabling hands-free automatic opening and closing of vehicle doors from ToF depth streams and point-cloud input.
- Achieved 97.2% recognition accuracy / 33 ms end-to-end latency under varying outdoor lighting conditions.

### PROJECTS

#### Legal QA System with Retrieval-Augmented Generation (RAG)

11/2025 - 05/2026

- Built a retrieval-augmented generation (RAG) system for legal question answering using LLMs and vector search. Indexed legal documents using dense embeddings and implemented semantic retrieval with FAISS. Integrated retrieved context with LLMs to generate accurate and grounded responses. Improved answer relevance and factual consistency compared to standalone LLM baselines.

#### LLM Post-Training with LoRA on Qwen for Instruction Tuning

08/2025 - 03/2026

- Applied LoRA-based fine-tuning on Qwen-7B for instruction-following tasks (QA, summarization, reasoning). Built training pipelines with Hugging Face and PEFT, reducing trainable parameters by over 99%. Tuned key LoRA hyperparameters (rank, alpha, dropout) to optimize performance-efficiency trade-offs under limited GPU resources. Improved instruction-following quality while reducing training cost and memory usage.

### EDUCATION

#### Michigan Technological University

Houghton, Michigan, USA

PhD in Computational Science and Engineering

05/2024 - 05/2027 Expected

#### Michigan Technological University

Houghton, Michigan, USA

Master in Computer Science

01/2022 - 04/2024

#### Shenzhen University

Shenzhen, Guangdong, China

Master in Optical Engineering

09/2016 - 06/2019

#### Nanchang University

Nanchang, Jiangxi, China

Bachelor in Optical Engineering

09/2012 - 07/2016